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Supplementary Material for "The Guarantee Holds and the Task Fails: Absorbing Stalls of Learned Policies under CBF Filters"

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1. Stall classification per policy (Experiment 7)

Table 1 gives the per-policy shares summarised in Figure 3(b) of the main text.

Table 1: Experiment 7, deterministic arm: shares of the n stalled states. Trap: no admissible command moves the body at the current heading. Position trap: no heading at the position admits motion. Rotation: an admissible rotation with $|\omega| \geq 0.5\,\omega_{\max}$ exists. QP: the exact weighted projection agrees with the filter. Law: the state arose on the intercept-law branch. Self-stop: share of immobile steps on which the nominal itself was zero.

Policy	n	Exact tier	Trap	Position trap	Rotation	QP	Law	Self-stop
BASE seed 1	74	1.00	1.00	0.00	1.00	1.00	1.00	0.41
BASE seed 2	39	1.00	0.97	0.00	1.00	0.97	0.97	0.66
BASE seed 3	42	1.00	1.00	0.00	1.00	1.00	1.00	0.57
AWARE	107	1.00	0.96	0.00	1.00	0.96	0.57	0.00

2. Names used in the paper and repository identifiers

The experiments were run and archived under internal identifiers, which the released records, checkpoints and pre-registration documents keep unchanged. Tables 2 and 3 map the names used in the paper to those identifiers. Record paths are relative to the repository's `results/` directory; each record names the commit it was produced at and its seed stream.

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Table 2: Policies, remedies and opponents.

Name in the paper	Repository identifier	Where it appears
BASE seeds 1, 2, 3	`BASE-60`, `BASE-61`, `BASE-62`	Experiments 3, 4, 7, 8, 9
OBS seeds 1, 2, 3	`OBS-60/61/62`	Experiments 3, 4
HOR seeds 1, 2, 3	`GAM-60/61/62`	Experiments 3, 4
OBSHOR seeds 1, 2, 3	`OBSGAM-60/61/62`	Experiment 4
penalty family (stall penalty)	`PEN-60/61/62`	Experiment 3
memory family (recurrent)	`MEM-60/61/62`	Experiment 3
rotation-only recovery	`REC` on the BASE seeds	Experiment 3
forward-offering recovery, parameter N	`RECF-`N; arms `RECF-BASE/OBS/OBSGAM-6x`	Experiments 4, 8, 9
perturbation baseline	`PERT-20`	Experiment 8
predictive filter	`PRED-PLAIN`, `PRED-RECF-20`	Experiment 9
FREE	`STR14`	Experiments 1, 8
AWARE	`RC19`	Experiments 1, 2, 7, 8, 9
anchored fine-tune of AWARE	`FT-B-41`	Experiment 2
three remedies of Experiment 1	`RA`, `RB`, `RC` (seeds 19–21)	Experiment 1
three penalty schedules	`DVARY`, `SPARSE`, `DRARE` (`-BASE`, `-PEN`, seeds 70–72)	Experiment 5
line law, planner law	`LINE`, `PLAN`	Experiment 6
greedy evader (G)	`GREEDY`	all
wall-hugging evader (H)	`HUG7` (`HUG`)	Experiments 3, 4, 6
learned attacker (A)	`ATK4-45`	Experiments 2–4, 6–9
safety hypothesis	`H-SAFE`	all

Table 3: Experiments, internal round identifiers, frozen commits, seed streams and records.

Exp.	Round	Frozen commit	Seed stream	Records
1	S4, S5	`9ea99f2`, `9a73105`	1 090 000+	`derisk/s4_cell_*.json`, `derisk/s5_arm_*.json`
2	S15	`7fcf9ba`	1 710 000+	`derisk/s15_round.json`, `derisk/s15_cells/`
3	S16	`32caa57`	1 720 000+	`derisk/s16_round.json`, `derisk/s16_arm_*.json`
4	S17	`d1ed3aa`	1 740 000+	`derisk/s17_round.json`, `derisk/s17_arm_*.json`
5	S18	`0ed69e6`	1 900 000+	`derisk/s18_round.json`, `derisk/s18_arm_*.json`
6	S19	`a252bc8`	2 000 000+	`derisk/s19_round.json`, `derisk/s19_cells/`
7	S26 B1	`b535af6`	3 000 000+	`derisk/s26_b1_round.json`, `derisk/s26_b1_cells/`
8	S26 B2	`b535af6`	1 740 000+	`derisk/s26_b2_round.json`, `derisk/s26_b2_cells/`
9	S26 B3, B4	`b535af6`	1 740 000+	same files as Experiment 8 (`PRED-*` cells)